

Mohammed Tawshif Hossain

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About

I am Mohammed Tawshif Hossain, recently have completed my undergraduate degree in Electrical and Electronic Engineering from Chittagong University of Engineering and Technology (CUET). I have strong passion for research, engineering and technology, with a particular focus on VLSI, meta optics, nano-photonics, Chip Designing, neutral-atom quantum computing, computational optics, biomedical optics and the emerging applications of quantum science and technology.

Driven by curiosity and a strong research mindset, I enjoy exploring interdisciplinary domains and actively seek opportunities to collaborate on innovative projects. Outside of academics and research, I enjoy playing football, watching football and engaging in intellectually stimulating discussions.

Research Interest

- Quantum photonics
- VLSI

- Meta Optics
- Integrated Photonics

- Quantum Technologies
- IC Designing

Education

Chittagong University of Engineering and Technology April 2022 – May 2026
B.Sc. in Electrical and Electronic Engineering

- GPA: 3.52 / 4.00

Chattogram College, Higher Secondary Certificate (HSC) June 2018 – January 2021

- GPA: 5.00 / 5.00

Chattogram Collegiate School, Secondary School Certificate (SSC) 2013 – 2018

- GPA: 5.00 / 5.00

Research Experiences

Ongoing Research Projects

Currently Working On June 2025 – Ongoing
Proposal: Reconfigurable on-chip optical tweezers and beam splitters using metasurfaces and phase-change materials.

- Target: Implementation of chip-scale tunability from optical tweezer to beam splitters.
- Target: Investigation of phase-change materials (GST, GSST, VO₂, In₃SbTe₂, Sb₂S₃, Sb₂Se₃).
- Target: Evaluation of temperature effects, optical power losses, CMOS compatibility, and tunability mechanisms (electrical, optical, and laser-induced).
- **Supervisor:** [Dr. Mehdi Hasan Chowdhury](#), Professor, Department of Electrical and Electronic Engineering, Chittagong University of Engineering & Technology (CUET), Chittagong, Bangladesh.

Undergraduate Thesis

June 2025 – Ongoing

Thesis Title: MultiFocus Dielectric Metasurface Optical Tweezers with Tailored Intensity Ratios for Parallel Nanoparticle Trapping

- Target: Implementation of selective optical trapping based on different intensity and optical force.
- Target: Investigation of adjoint and optimization methods for ensuring specific optical force and intensity at specific positions with multiple trapping capabilities.
- Target: Analyzation of optical force effect on different particles.
- **Supervisor:** [Dr. Mehdi Hasan Chowdhury](#), Professor, Department of Electrical and Electronic Engineering,

Past Research Projects

Decoding Memory with Explainable AI: A Large-Scale EEG-Based Machine Learning Study of Encoding vs. Retrieval [First Author] [Details] Feb. 2024 – June 2025

- Implementation of machine learning techniques on EEG data from the Penn Electrophysiology of Encoding and Retrieval Study (PEERS) [Details], involving 100 participants across more than 400 sessions.
- Utilization of Discrete Wavelet Transform (DWT) to analyze EEG signals from six critical brain regions during memory encoding and retrieval.
- Application of Explainable AI (XAI) methods, specifically SHapley Additive exPlanations (SHAP), to interpret neurophysiological markers of memory processes.

Research Project July 2024 – Dec. 2024
Spectrogram-driven Emotion Detection from Electroencephalogram [Details]

- Employed Short-Time Fourier Transform (STFT) to convert EEG signals into time-frequency spectrograms for CNN-based emotion recognition.
- Achieved 99.80% classification accuracy for emotional states (positive, negative, neutral) using processed spectrogram representations.
- Demonstrated reduced performance (66%) when using raw spectrograms, highlighting the importance of preprocessing.
- **Supervisor:** Md. Fazlul Karim Khondakar, Assistant Professor, Department of Biomedical Engineering, CUET.

Industrial Attachment (Two Weeks)

Biomedical Engineering Industrial Training [GE Healthcare] Two Weeks

- Hands-on exposure to advanced medical imaging systems, including **GE CT scanners**, focusing on gantry structure, X-ray tube units, power distribution units (PDU), high-voltage components, and cooling systems.
- Practical understanding of **cyclotron and PET radiopharmacy facilities**, including radiochemistry hotcells, automated synthesis modules (**GE TRACERlab and FASTlab**), target gas handling, HPLC purification units, and radiation safety protocols.
- Study of **MRI (1.5T) systems**, covering superconducting magnets, gradient and RF coils, cryogenic helium compressors, RF penetration panels, and preventive maintenance (PM) procedures.
- Familiarization with **ultrasound imaging systems (GE Voluson E10)**, including probe technologies, 4D imaging capabilities, clinical applications, and routine preventive maintenance.
- Exposure to **Exercise Tolerance Test (ETT) systems**, including GE T2100-ST1 medical treadmills, ECG-based cardiac stress testing workflows, and patient safety considerations.

Academic Projects

Automatic Power Factor Corrector (APFC) & Tier-3 Data Center Electrical Design [Details]

- **Electrical Infrastructure:** Designed complete Tier-3 Data Center systems including MDB, EMDB, SDB, ESDB, and detailed Single Line Diagrams (SLD).
- **Load Analysis & Engineering:** Performed detailed load analysis for Critical (UPS, CRAC, Server Racks) and Normal utility loads; calculated breaker sizing and cable selection.
- **Power Quality Optimization:** Engineered a 30 kVAR Detuned APFC system (7% reactor) to improve Power Factor from 0.8 to 0.99, mitigating harmonics and kVA penalties.
- **Tools Used:** AutoCAD Electrical, ETAP.
- **Standards Followed:** IEC 60364, IEEE 519.

2-Bit Full Adder Design and Physical Implementation [Details]

- **System Design & Functional Analysis:** Defined functional specifications, I/O requirements, architectural design, and developed state diagrams for a 2-bit Full Adder.

- Logic Design: Created gate-level schematic and standard-cell symbols; verified combinational logic functionality through simulation.
- Circuit Design & Layout: Designed transistor-level CMOS implementation and developed custom standard-cell layouts following VLSI design rules.
- Physical Verification: Performed layout placement, routing, Layout Versus Schematic (LVS) verification, and Design Rule Check (DRC) validation.
- Chip Integration: Completed top-level chip layout with I/O pad placement and conducted post-layout simulation to verify functionality and parasitic effects.
- Tools Used: Cadence Virtuoso (Schematic, Layout, LVS, DRC, Post-Layout Verification).

EEG Signal Processing With Python MNE

- EEG Signal: Signal Data Collection, Signal Pre-Processing, Signal processing (Features extraction using STFT, Wavelet Transform), Feature Engineering and Selection (PCA, Correlation Map, Statistical t-test)
- Deployment of the Machine Learning Model on the processed Dataset and Interpretation of the results.
- Tools Used: Python, Jupyter Notebook, Kaggle, Google Colab.
- Library Used: Python MNE, Pandas, Numpy, Tensorflow, Matplotlib, Seaborn.

[Coursera: Unsupervised Learning]

A deep Q-learning neural network in order to land a virtual lunar lander on Mars.

- State-Action Value Function, bellman Equation
- Improved Neural Network Architecture
- Tools Used: Jupyter notebook.

Classifying diseases on chest x-rays using a neural network.

- Image Processing on chest x-rays data (Gray Scaling, Masking), Augmentation of the Images, Mitigation of Class Imbalance in Datasets.
- Labeling of the images, Implementation of the Deep learning Models.
- Tools Used: Python, Jupyter Notebook
- Library Used: Scikit-image, SciPy

Design and Implementation of Gas Leakage and Fire Alarm System [\[Details\]](#)

- Designed and implemented a Gas Leakage and Fire Alarm System using Arduino and MQ5 gas sensors.
- Developed real-time monitoring and alert mechanisms with visual (LCD display) and audible alarms for early detection of gas leaks and fire hazards.
- Tools Used: Arduino Uno, C++

Digital Thermometer Construction [\[Details\]](#)

- Implementation of micro controller core processor for the digital thermometer managing input from sensors and controlling output to displays.
- Temperature Sensing with Thermistor: Utilization of a KTY81 PTC thermistor, in which resistance increases with temperature.
- Tools Used: PIC16F877A microcontroller, KTY81 Thermistor, Proteus

500 KVA DISTRIBUTION TRANSFORMER DESIGN

- Designed a 500 kVA, 11 kV/415 V, delta/star transformer with tapings, cooling method, and efficiency/regulation calculations.
- Evaluated efficiency and voltage regulation at various loads and power factors for operational performance.
- Tools Used: Matlab.

Skills

Computer Skills:

Programming Languages: C++, MATLAB, Python, Git, Linux

Software: MATLAB, Cadence Virtuoso, Comsol Multiphysics, Lumerical FDTD, Jupyter Notebook, Visual Studio Code, AWR, ADS, Proteus

Typography: LaTeX, MS Word, MS PowerPoint.

Publications

Journal Article (In press)

- **Decoding Memory with Explainable AI: A Large-Scale EEG-Based Machine Learning Study of Encoding vs. Retrieval**

Status: In Press [\[Details\]](#)

- **Mohammed Tawshif Hossain**, Adnan Sami Sarkar, Arnab Chowdhury, Rajesh Mitra, Raiyan Rahman, Mahdy Chowdhury

Conference Paper (In Press) [\[Details\]](#)

- Md. Taisirul Muktadi, Tanzeem Tahmeed Reza, **Mohammed Tawshif Hossain**, Eshat Jahan Arifa, Md. Fazlul Karim Khondakar.

Spectrogram-Driven Emotion Detection from Electroencephalogram.

In Proceedings of the 2025 4th International Conference on Electrical, Computer and Communication Engineering (ECCE),

13–15 February 2025, CUET, Chittagong

Competition

IEEE Signal Processing Cup

Aug 2024

Deepfake Face Detection In The Wild Competition [\[Details\]](#)

- **Handling 30+ GB imbalanced deepfake dataset:** Strategically split 219,470 fake images into four subsets while keeping all 42,690 real images, enabling efficient training with limited resources (Kaggle T4x2 GPU).
- **Image augmentation to mitigate overfitting and balance classes:** Applied random horizontal/vertical flips, brightness (± 0.1), and contrast ($[0.8, 1.2]$) to real images, synthetically expanding the dataset and reducing bias toward the majority fake class.
- **Deep CNN implementation with strong generalization:** Designed a 4-block CNN (32→256 filters, batch norm, dropout 30–50%) that achieved 97.59% validation accuracy and accuracy of nearly 85% in test sets demonstrating robust performance across four training phases.

Online Courses

[\[Details\]](#)

Ongoing

Online Optics and Photonics Course

- **Maxwell's Equations & Wave Theory:** In-depth understanding of Maxwell's equations, including integral and differential forms, physical interpretations, EM wave equation derivation, and detailed analysis of wave propagation, boundary conditions, and antenna radiation.
- **Optical Force, Waveguides & Antennas:** Study of light momentum, energy conservation, Maxwell stress tensor, plasmonics, surface plasmons, rectangular and cylindrical waveguides, TE & TM modes, Rayleigh scattering, and antenna theories with detailed mathematical modeling.
- **Simulation & Practical Training:** Hands-on training with **COMSOL Multiphysics** and **Lumerical FDTD simulation tools**, including 2D and 3D modeling of dielectric, plasmonic, and chiral objects, oblique incident light setups, and the use of simulation software for advanced optical designs and analysis.

[\[Details\]](#)

Ongoing

Online Quantum Mechanics and Quantum Computing Course

- **Quantum Mechanics Fundamentals:** Wave-particle duality, quantization, quantum tunneling, and the

mathematical formalism of QM.

- Quantum Computing Basics: Qubits, superposition, entanglement, and basic quantum circuits.
- Simulation & Practical Training: Hands-on training with **COMSOL Multiphysics** for quantum system designs and analysis.

Machine Learning Specialization by Andrew Ng (Coursera)

Completed

- **Classical ML with NumPy & scikit-learn:** Built and validated supervised models for regression and binary classification, including linear and logistic regression.
- **Neural Networks & Deep Learning:** Designed, trained, and evaluated neural networks in TensorFlow for multi-class classification.
- **Decision Trees & Ensembles:** Implemented decision trees and ensemble methods (e.g., Random Forests, Gradient Boosting) for robust predictive modeling.
- **Unsupervised Learning:** Applied clustering and anomaly detection to uncover patterns in unlabeled data.
- **Recommender Systems:** Developed collaborative filtering and content-based deep learning recommenders.
- **Deep Reinforcement Learning:** Built and trained deep RL agents for sequential decision-making.
- **ML Best Practices:** Model validation, hyperparameter tuning, reproducibility, and deployment-ready workflows.